

Skills Summary

Factor Detective: Factors, Multiples, and Patterns

Use this reference while completing your worksheet or when studying.

1. Factor pairs, primes and composites

A **factor pair** is two numbers that multiply to make the target. Hunt for them in order — 1, 2, 3, 4... — and stop when the two numbers in a pair meet.

Prime: exactly two factors, 1 and itself. **Composite:** more than two. (1 is neither.)

Example: Complete the factor pair $3 \times \underline{\quad} = 18$, and say whether 18 is prime.

Step 1. A factor pair asks how many equal groups fit exactly, so this is a division question in disguise.

Step 2. Count by threes: 3, 6, 9, 12, 15, 18 — six threes, so $18 \div 3 = 6$.

The partner factor is **6**, and since 18 has the extra pair 3×6 , it is composite.

Try it: Complete $7 \times \underline{\quad} = 56$.

check: 8

2. Divisibility rules: test without dividing

Divisible by **2** if the last digit is even; by **5** if it ends in 0 or 5; by **10** if it ends in 0.

Divisible by **3** if the *sum of the digits* is divisible by 3 — and by **9** if that digit sum is divisible by 9.

Example: Is 258 divisible by 3?

Step 1. Dividing would work, but the digit-sum rule answers it in one line, so add the digits first.

Step 2. $2 + 5 + 8 = 15$, and $15 = 3 \times 5$ is divisible by 3.

The digit sum is **15**, so 258 *is* divisible by 3.

Try it: Add the digits of 471 to test it for 3.

check: digit sum 12, so yes

3. Multiples and the first common multiple

The **multiples** of a number are its skip-count: 4, 8, 12, 16 . . . Every number is a multiple of each of its factors.

To find the **first common multiple** of two numbers, skip-count both and stop at the first number that appears in both lists.

Example: Find the first common multiple of 4 and 6.

Step 1. Two skip-counts have to land on the same number, so list them side by side and watch for the first match.

Step 2. Fours: 4, 8, 12. Sixes: 6, 12. The first match is the third multiple of 4.

$$4 \times 3 = 12$$

Try it: Find the first common multiple of 3 and 5.

check: 15

4. Number patterns: find the step, then jump

Subtract two neighbours to find the **step**. Then remember that reaching the n th number takes $n - 1$ steps, because the first number is already there.

$$n\text{th number} = \text{first number} + (\text{number of jumps}) \times \text{step}.$$

Example: A pattern starts at 5 and adds 6 each time. What is the 7th number?

Step 1. Writing out seven terms works, but counting jumps is quicker and does not drift — so find how many jumps are needed first.

Step 2. Reaching the 7th number takes 6 jumps: $5 + 6 \times 6 = 5 + 36$.

The 7th number is **41**.

Try it: A pattern starts at 2 and adds 9 each time. Find the 6th number.

check: 47